

# **EduGenie – AI Integration**

## **Project: EduGenie – Google Gemini Powered Learning Assistant**

### **1. Introduction**

EduGenie uses the Google Gemini API to provide AI-powered learning assistance. The AI service processes student requests and generates useful educational responses.

### **2. AI Service**

**AI Service:** Google Gemini API

The Gemini API is connected to the EduGenie backend and is used for AI-based learning tasks.

### **3. API Integration Process**

The integration works as follows:

Student Input



EduGenie Frontend



FastAPI Backend



Learning Module



Google Gemini API



AI Generated Response



FastAPI Backend



EduGenie Frontend



Student

## **4. AI-Powered Features**

### **4.1 Question Answering**

The AI processes academic questions and generates clear answers.

### **4.2 Simplified Explanation**

The AI helps explain educational concepts in simpler language.

### **4.3 Quiz Generation**

The AI generates quiz questions based on the requested topic.

### **4.4 Text Summarization**

The AI processes educational text and produces a concise summary.

### **4.5 Learning Recommendations**

The AI provides recommendations to help students decide what they can learn next.

## **5. API Key Security**

The Gemini API key should be stored as an environment variable rather than directly inside the source code.

This helps prevent sensitive credentials from being exposed in the public GitHub repository.

## **6. Error Handling**

If the AI service is unavailable or the API configuration is incorrect, the application should handle the error and provide an understandable message to the user.

## **7. Conclusion**

Google Gemini provides the

AI processing capability of EduGenie and enables the application to deliver multiple AI-powered learning features through the FastAPI backend.